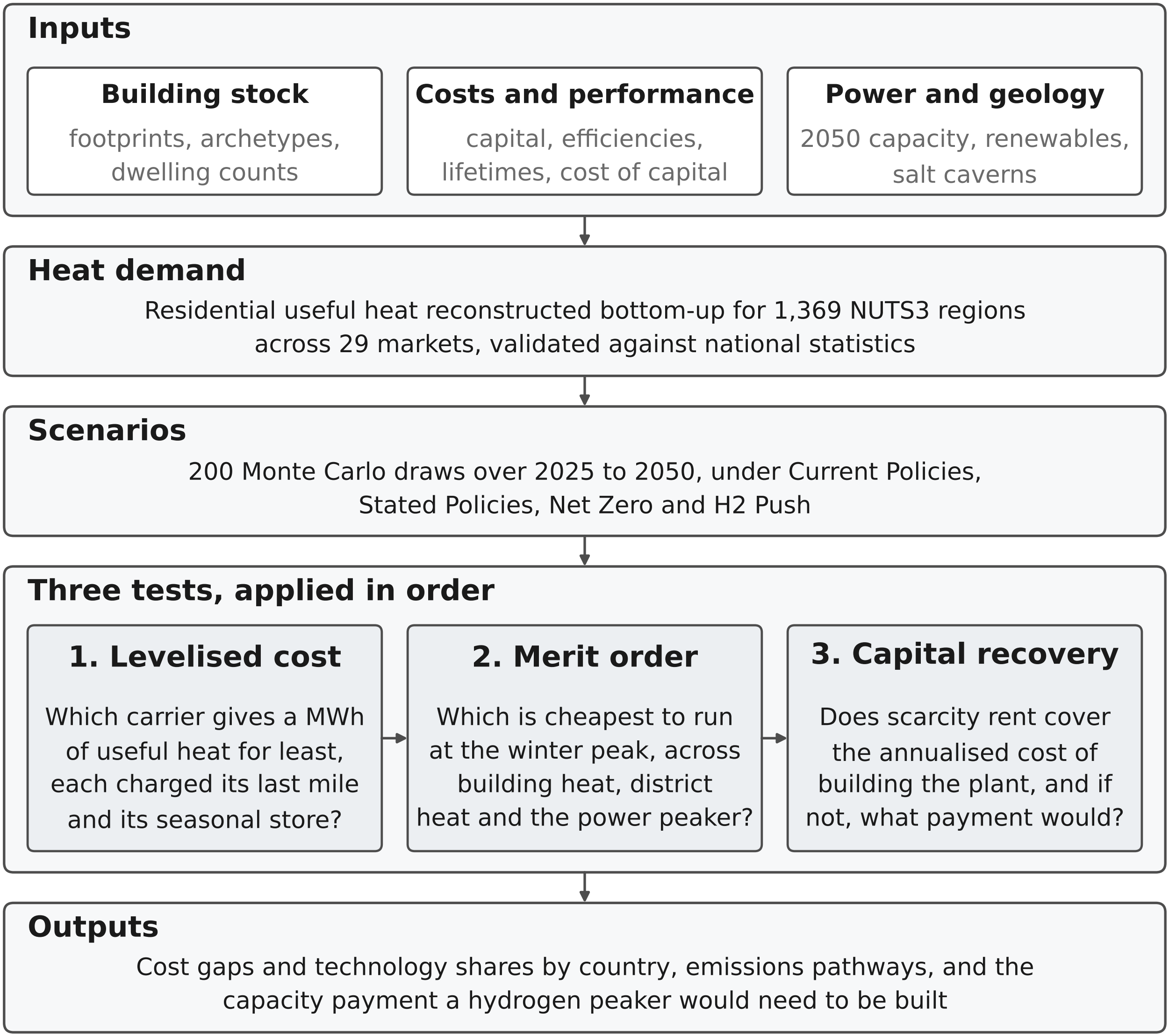
Inputs
Building stock
footprints, archetypes, dwelling counts
Costs and performance
capital, efficiencies, lifetimes, cost of capital
Power and geology
2050 capacity, renewables, salt caverns
Heat demand
Residential useful heat reconstructed bottom-up for 1,369 NUTS3 regions across 29 markets, validated against national statistics
Scenarios
200 Monte Carlo draws over 2025 to 2050, under Current Policies, Stated Policies, Net Zero and H2 Push
Three tests, applied in order
1. Levelised cost
Which carrier gives a MWh of useful heat for least, each charged its last mile and its seasonal store?
2. Merit order
Which is cheapest to run at the winter peak, across building heat, district heat and the power peaker?
3. Capital recovery
Does scarcity rent cover the annualised cost of building the plant, and if not, what payment would?
Outputs
Cost gaps and technology shares by country, emissions pathways, and the capacity payment a hydrogen peaker would need to be built